Marouane BENHASSI

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Place and Date of Birth Marrakesh, 12th September 1997

Bio. After obtaining my master degree in high energy physics, astrophysics and computational physics, I am currently a part-time teacher in physics at Massar Al Omam high school, I am very passionate about particle physics, detectors and improving my coding skills. I am interested in simulation and data analysis using ROOT CERN, Python and C++.

EDUCATION

2022 – Present

IFJ PAN PPSS School 2022, Kraków, Poland

The first aim is to introduce basics of the Standard Model of particle physics, methodology of research in this field and statistical data analysis. The second aim is to give the opportunity for students to participate in a real scientific research.

- Title of the research project: New physics searches in ultra-peripheral collisions in ALICE

2020 – 2021

English Courses

Center for Language and Culture (CLC), Marrakesh, Morocco

2019 – 2021

Master degree in hight energy physics, astrophysics and computational physics

High Energy and Astrophysics Laboratory, Semlalia Faculty of Sciences, Cadi Ayyad University, Marrakesh, Morocco

- **Rank** : 1st in the M2
- Title of Master Degree Thesis: Neutrinos physics with KM3NeT detector

2015 – 2019

Bachelor degree in modern physics

Semlalia Faculty of Sciences, Cadi Ayyad University, Marrakesh, Morocco

- Title of Bachelor Degree Thesis: Particle matter interaction and multi-detectors

PROFESSIONAL EXPERIENCES

2022 – Present

Research project, IFJ PAN, Poland, one month

- Supervisor: Adam Matyja
- Title: New physics searches in ultra-peripheral collisions in ALICE
- Description: Perform ROOT based analysis of ALICE data coming from the UPCs of Pb-Pb at $\sqrt{s_{NN}}$ = 5.02 TeV during Run 2. Comparison between Data and Monte Carlo predictions.

2021 – Present

Part-time teacher in physics, Massar Al Omam high school, Marrakesh, Morocco

- Description : Teaching physics for high school and middle school students

2020 – 2021

Internship at ESMaR laboratory, Rabat, Morocco, three months

- Supervisor: Jihad Boumaaza
- Description: Construction of the KM3NeT's detector used to detect Cherenkov radiations emitted by charged particles during their propagation in the Mediterranean Sea.

2020 – 2021 **Master thesis, Marrakesh, Morocco, four months**

- Title : Neutrino physics with the KM3NeT detector
- Supervisor : Mohamed Chabab
- Description : Study of neutrino oscillations and simulation of data from the ORCA detector (One of KM3NeT's detectors) to obtain the oscillation parameters using the monte carlo simulation
- The report is available at : https://www.researchgate.net/publication/354640423_Neutrino_physics_with_the_KM3NeT_detector

2018 – 2019 **Bachelor thesis, Marrakesh, Morocco**

- Title : Particle matter interaction and multi-detectors
- Supervisor : Driss Goujdami
- Description : Study of the interactions of heavy and light charged particles with the matter, the interaction of radiation-matter, and explanation of the ATLAS and CMS experiments
- The report is available at : https://www.researchgate.net/publication/348842906_Interaction_particules_matiere_et_multi-detecteurs

PROJECTS

2021 – 2022 **Project: Perform a typical analysis in particle physics**

- Description : having a model of signal and background (in the form of generated "Monte Carlo" events – MC samples), figure out a selection criteria that can be applied on data events to suppress the background contribution.

2021 – 2022 **Project : Analysis the data of the Z boson decay**

- Description : Analyze data from the Z boson decay into two leptons using ROOT CERN software.

2019 – 2020 **Project : Analysis the data of the Higgs boson decay**

- Description : Analyze the data of the Higgs boson decay into two photons using ROOT CERN software.

2019 – 2020 **Project : Verification of the central limit theorem**

- Description : Verification of the central limit theorem (CLT) using the ROOT CERN software.

SCHOOLS AND WORKSHOPS

2021 – 2022 **Coache at the Django Girls workshop, CERN, Switzerland**

- Description : The goal of Django Girls is to bring new people into technology, especially women who are under-represented in this field.

2021 – 2022 **First Pan-African Astro-Particle and Collider Physics Workshop, Rabat, Morocco, one week**

- Description : The intent of the workshop is to provide a forum for exchange of information among African researchers in the fields of particle physics, Astro-Particle physics and Collider Particle Physics.

2020 – 2021 **Summer School on Methods of Effective Field Theory and Lattice Field Theory, Bad Honnef, Germany, two weeks**

- Description : This training provide young scientists with the tools for tackling the most interesting and still open problems in particle, astrophysics and nuclear physics.

2020 – 2021 **ICTP Summer School on Particle Physics, Trieste, Italy, two weeks**

- Description : Acquire a detailed overview of particle physics from the basics of Standard Model phenomenology to the most important areas where significant progress has been achieved recently.

2020 – 2021 **PyHEP 2021 Workshop, one week**

- Description : The PyHEP workshops are a series of workshops initiated and supported by the HEP Software Foundation (HSF) with the aim to provide an environment to discuss and promote the usage of Python in the HEP community at large.

SKILLS

General skills

- Ability to collaborate in a team, in an international environment within a team multidisciplinary
- Ability to present results in international congresses, workshops and meetings with MS Power Point
- Problem solving
- Perseverance and curiosity

Languages

- Arabic : Maternel language
- English : Good
- French : Advanced

Computer skills

- Operating system : GNU/Linux and Microsoft Windows
- Programing Language : **Experienced** : C++, Python, MATLAB **Familiar** : Fortran, Maple, Mathematica, HTML, CSS
- Framework and Libraries : ROOT CERN, Geant4 (biginner)
- Technologies : LaTeX, Microsoft office pack, Monte Carlo Simulation (biginner)

ADDITIONAL INFORMATION

2020 – present Member of HEPA Club: Making physics simple for everyone, Marrakesh

Hobbies Reading, Playing guitar, Traveling, Cooking, Playing football

REFERENCES

Adam Matyja

- Member of ALIC experiment and the Institute of Nuclear Physics Polish Academy of Sciences (IFJ PAN)
- Researcher
- My supervisor
- adam.tomasz.matyja@cern.ch

Mohamed EL KACIMI

- Member of ATLAS experiment and the High Energy and Astrophysics Laboratory
- Senior university teacher
- My Master teacher
- elkacimi@uca.ac.ma